PRE-ALGEBRA (8TH) · UNIT 4: SYSTEMS OF LINEAR EQUATIONS

Food Truck Face-Off

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students compare a taco truck and a burger truck for a festival spot. Cost lines, customer orders, and profit equations all lead to systems that students solve and interpret.

Every method (graphing, substitution, elimination) and every solution type appears in a business context.

Standards Addressed

- 8.EE.C.8a Solutions of a system correspond to intersection points
- 8.EE.C.8b Solve systems algebraically and by inspection
- 8.EE.C.8c Real-world problems with two linear equations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the truck cards. Part 1: cost lines and break-even.
Day 2	Part 2: customer orders and elimination; special systems.
Day 3	Part 3: profit comparisons and the business report.

Materials

- Student handout with truck cards
- Graph paper
- Calculators
- Optional: menu boards from local food trucks

How the Three Versions Differ

Level	What changes
CP2	Tables and substitution with starters, one elimination system, and a profit comparison at 50 items.
CP1	All three methods, break-even points, special systems, and a profit inequality.
Honors	Three concurrent lines, a three-item order, parameters for the number of solutions, and a designed truck.

Watch For (Common Misconceptions)

- Forgetting to multiply every term when scaling an equation for elimination.
- Stopping after finding one variable.
- Calling parallel lines "infinitely many solutions."

Extension Ideas

- Graph profit lines and shade where each truck makes money.
- Survey classmates about prices and adjust the sales prediction.
- Add a weekend discount and resolve the systems.

Answer Key – CP2

Part 1: Cost Lines

#	Task	Answer
1	Write a cost equation for each truck.	Taco: $y = x + 120$. Burger: $y = 2x + 80$.
2	Complete a table for $x = 0, 20, 40, 60$. Where are the costs equal?	Taco: 120, 140, 160, 180. Burger: 80, 120, 160, 200. Equal at $x = 40$ items, \$160.
3	Solve the system by substitution to check: $x + 120 = 2x + 80$.	$x = 40$; $y = 40 + 120 = 160$. The solution is (40, 160).

Part 2: Customer Orders

#	Task	Answer
4	Write a system for the number of tacos t and burgers b.	$t + b = 10$ and $3t + 5b = 38$.
5	Solve the system by substitution.	$b = 10 - t \rightarrow 3t + 5(10 - t) = 38 \rightarrow 50 - 2t = 38 \rightarrow t = 6, b = 4$.
6	Check your answer in both equations.	$6 + 4 = 10$ ✓ and $3(6) + 5(4) = 18 + 20 = 38$ ✓.

Part 3: Elimination & Profit

#	Task	Answer
7	Solve by elimination: $2t + 3b = 21$ and $2t + b = 11$.	Subtract: $2b = 10 \rightarrow b = 5$; $2t + 5 = 11 \rightarrow t = 3$.
8	Each truck sells 50 items. Find each truck's profit (sales minus costs).	Taco: $3(50) - (120 + 50) = -\$20$. Burger: $5(50) - (80 + 100) = \$70$.
9	Which truck would you choose for the festival? Use your numbers.	Answers vary; the burger truck earns more when each sells 50 items.

Reflection

#	Prompt	Look for
R1	What does the point (40, 160) mean for the two trucks?	If each truck makes 40 items, both have the same cost: \$160.
R2	How do you check a solution to a system?	Substitute the values into both equations; both must be true.

Answer Key – CP1

Part 1: Costs & Break-Even

#	Task	Answer
1	Solve the cost system $y = x + 120$ and $y = 2x + 80$ by substitution, and interpret the solution.	(40, 160): at 40 items both trucks spend $160.
2	Find each truck's break-even point (revenue = cost).	Taco: $3x = x + 120 \rightarrow x = 60$ tacos. Burger: $5x = 2x + 80 \rightarrow x \approx 26.7$, so 27 burgers.
3	Graph both cost lines. For how many items is the taco truck's cost lower?	For more than 40 items (its line has the smaller slope).

Part 2: Orders & Elimination

#	Task	Answer
4	A group buys 14 items for $54 (tacos $3, burgers $5). How many of each?	$t + b = 14$, $3t + 5b = 54 \rightarrow 3t + 70 - 5t = 54 \rightarrow t = 8$, $b = 6$.
5	Solve by elimination: $4t + 3b = 39$ and $2t + 5b = 37$.	Double the second: $4t + 10b = 74$. Subtract: $7b = 35 \rightarrow b = 5$; $4t + 15 = 39 \rightarrow t = 6$.
6	How many solutions does each system have? (a) $y = 3x + 2$ and $6x - 2y = -4$ (b) $y = 2x + 1$ and $y = 2x - 5$	(a) Infinitely many: the second equation is $y = 3x + 2$. (b) No solution: same slope, different intercepts.

Part 3: Profit Showdown

#	Task	Answer
7	Write each truck's profit equation and decide whether the profits are ever equal for the same number of items.	Taco: $P = 2x - 120$. Burger: $P = 3x - 80$. Burger − taco $= x + 40 > 0$, so they are never equal; the burger truck always earns $40 + x more.
8	The taco truck sells 3 tacos for every burger the burger truck sells. If the burger truck sells b burgers, for which b does the taco truck earn more?	$2(3b) - 120 > 3b - 80 \rightarrow 3b > 40 \rightarrow b \geq 14$ burgers.
9	Write a recommendation to the committee using your results.	Answers vary; mention that the taco truck wins only if it sells about 3 times as many items (at least 42 tacos to 14 burgers).

Reflection

#	Prompt	Look for
R1	What do parallel lines mean in a business setting?	Two quantities that grow at the same rate with different starting values never become equal.
R2	When is elimination faster than substitution?	When a variable already has matching or opposite coefficients, or can easily be made to.

Answer Key – Honors

Part 1: Three Trucks

#	Task	Answer
1	Find where each pair of cost lines meets.	All three pairs meet at the same point: $x + 120 = 2x + 80 \rightarrow x = 40$; $1.5(40) + 100 = 160$. Every pair meets at (40, 160).
2	Which truck has the lowest cost for fewer than 40 items? For more than 40 items?	Fewer than 40: burger truck (lowest rent). More than 40: taco truck (lowest cost per item).
3	Explain why three lines with different slopes can meet in one point.	Each pair has exactly one solution; here those solutions happen to be the same point, so the lines are concurrent.

Part 2: Festival Orders

#	Task	Answer
4	A school buys 100 items for $330, with as many drinks as tacos. How many of each did they buy?	$d = t$; $2t + b = 100$; $5t + 5b = 330 \rightarrow t + b = 66$. Subtract: $t = 34$, $b = 32$, $d = 34$. Check: $34 + 32 + 34 = 100$ and $102 + 160 + 68 = 330$.
5	For which k does $2x + 3y = 12$ and $4x + ky = 24$ have infinitely many solutions?	$k = 6$ (the second equation is 2 times the first).
6	Explain why no value of k makes that system have no solution.	The constants already have the ratio $24 : 12 = 2$. Parallel lines would need $k = 6$, but then the lines coincide; any other k gives exactly one solution.

Part 3: Pricing Strategy

#	Task	Answer
7	The burger truck sells 50 burgers at $5. If it raises the price to $6, it expects to sell 20% fewer. Compare the profits.	$5: $250 - (80 + 100) = $70. $6: 40 burgers → $240 - (80 + 80) = $80. The price increase earns $10 more.
8	Design a third truck (rent, cost per item, price) that breaks even at exactly 30 items. Show the equation.	Answers vary. Example: rent $60, cost $2, price $4: $4x = 2x + 60 \rightarrow x = 30$.
9	Write a system whose solution shows when your truck's profit equals the burger truck's profit, and solve it.	Example: $P = 2x - 60$ and $P = 3x - 80 \rightarrow x = 20$ items, $P = -$$20 (both still losing money at 20 items).

Reflection

#	Prompt	Look for
R1	How can you predict the number of solutions from the slopes and intercepts?	Different slopes: one solution. Same slope, different intercept: none. Same slope and intercept: infinitely many.
R2	Why can raising a price increase profit even when fewer items sell?	Each sale earns more and fewer items mean lower supply cost, so profit can rise if sales do not drop too much.